Information and Communication Technology

Teacher's Guide

Grade 10

(Revised-2026)

Department of Information Technology
National Institute of Education

Maharagama

Sri Lanka

www.nie.ac.lk

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Message from the Director General

The first phase of the new competency based curriculum, with 8 years curriculum cycle was introduced to secondary education in Sri Lanka in 2007 replacing the existed content based education system with basic objective of developing the national level competencies recommended by the National Education Commission.

The second phase of the curriculum cycle to be introduced to grades 6 and 10 starts from 2015. For this purpose, National Institute of Education has introduced a rationalization process and developed rationalized syllabi for these grades using research based outcomes and various suggestions made by different stakeholders.

In the rationalization process, vertical integration has been used to systematically develop the competency levels in all subjects from fundamentals to advanced levels using the bottom up approach. Horizontal integration is used to minimize the overlapping in the subject content and to reduce the content over loading in the subjects to produce more students friendly and implementable curricular.

A new format has been introduced to the teachers' guide with the aim of providing the teachers with the required guidance in the areas of lesson planning, teaching, carrying out activities and measurement and evaluation. These guidelines will help the teachers to be more productive and effective in the classroom.

The new teachers' guides provide freedom to the teachers in selecting quality inputs and additional activities to develop the competencies of the students. The new teachers' guides are not loaded with subject content that is covered in the recommended textbooks. Therefore, it is essential for the teacher to use the new teachers' guides simultaneously with the relevant textbooks prepared by Education Publication Department as reference guides to be more aware of the syllabi.

The basic objectives of the rationalized syllabi and the new format of teachers' guide and newly developed textbooks are to bring a shift from the teacher centered education system into a student centered and more activity based education system in order to develop the competencies and skills of the school leavers and to enable the system to produce suitable human resource to the world of work.

I would like to take this opportunity to thank the members of Academic Affairs Board and Council of National Institute of Education and all the resource persons who have immensely contributed in developing these new teacher guides.

Director General
National Institute of Education

Message from the Deputy Director General

Education from the past has been constantly changing and forging forward. In recent years, these changes have become quite rapid. Past two decades have witnessed a high surge in teaching methodologies as well as in the use of technological tools and in the field of knowledge creation.

Accordingly, the National Institute of Education is in the process or taking appropriate and timely steps with regard to the education reforms of 2015.

It is with immense pleasure that this Teachers' Guide where the new curriculum has been planned based on a thorough study of the changes that have taken place in the global context adopted in terms of local needs based on a student-centered learning-teaching approach, is presented to you teachers who serve as the pilots of the schools system.

An instructional manual of this nature is provided to you with the confidence that, you will be able to make a greater contribution using this.

There is no doubt whatsoever that this Teachers' Guide will provide substantial support in the classroom teaching-learning process at the same time. Furthermore, the teacher will have better control of the classroom with a constructive approach in selecting modern resource materials and following guide lines given in this book.

I trust that through the careful study of this Teacher’s Guide provided to you, you will act with commitment in the generation of a greatly creative set of students capable of helping Sri Lanka move socially as well as economically forward.

This Teachers' Guide is the outcome of the expertise and unflagging commitment of a team of subject teachers and academics in the field of Education.

While expressing my sincere appreciation of this task performed for the development of the education system, my heartfelt thanks go to all of you who contributed your knowledge and skills in making this document such a landmark in the field.

Deputy Director General
Faculty of Science and Technology

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Instructions for use of the Teacher's Guide

This Teacher's Guide covers all competency levels of the Grade 10 ICT syllabus and is designed as relevant to the learning- teaching process. In addition to the competency levels and subject matter, some more related concepts and terms have also been highlighted. The objective is to provide the user with more in-depth understanding of the syllabus. The guidance for lesson plans may be modified by the teacher to suit the relevant context. The assessment and

evaluation guidance may also be modified according to the needs of the situation. The quality inputs mentioned in this case are specified websites that accompany this Teacher's Guide. These are meant for the use of both teachers and students and could be integrated in their learning –teaching process to their convenience.

Guidance is provided for the preparation of lab-sheets and it is very important to involve students in laboratory practicals as it provides them with better understanding of the subject matter and could lead them to develop skills necessary to ensure a better place in the world of work that demands ICT applications.

For laboratory practicals following items of software are recommended.

Operating Systems - a flavour of Linux

Office applications - a flavour of Open Office

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Information and Communication Technology

Syllabus

Grade 10

(Revised - 2026)

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1. Introduction

The Information and Communication Technology sector is acknowledged worldwide as a tool that could be used to increase the productivity, efficiency and effectiveness of work. However, in Sri Lanka, the level of ICT skills of the majority of the students is not adequate to meet the current requirements of business and industry. This is mainly due to lack of opportunities for students to study ICT related subjects in the school curriculum. The student should learn at school a wide variety of competencies for different needs of life in the changing world. They should have various views and different ways to continue studies and proceed to employment.

Since Sri Lanka is in the early stages of introducing ICT to the lower grades, the present syllabus does not demand any ICT knowledge as an entry requirement. Therefore, this syllabus is intended to introduce ICT as a technical subject to be offered at the G.C.E (O/L) Examination. The main objective of this syllabus is to develop the competencies to use ICT tools and to build a basic theoretical base for students to pursue higher studies in ICT.

The curriculum that was implemented in 2006 (Grade 10) has now been revised based on the evidence of a survey conducted, advice of the veterans in this discipline and developments in the field of ICT. The syllabus of Grade 10 has not undergone much revision but certain content that was best advised to be removed or toned down to alleviate cognitive burden of the target group. Subject matter on databases was streamlined to provide students with better view and involvement as this has more than a mere academic value and add values to the life-skills of students.

2. National Goals

1. Based on the concept of respecting human values and understanding the differences between the Sri Lankan multi-cultural society, building up the nation and confirming the identity of Sri Lanka by promoting national integrity, national unity, national coherence and peace.

2. While responding to the challenges of the dynamic world, identifying and conserving the National heritage. Creating an environment which comprises of the conventions of social justice and the democratic life to promote the characteristics of respecting the human rights, being aware of the responsibilities, concerning each other with affectionate relationships.

3. Promoting a sustainable life style based on the people's mental and physical well being and the concept of human values Promoting the positive feelings needed for balanced personality with the qualities of creative skills, initiative, critical thinking and being responsible.

4. Through education, developing the human resources, needed for the progress of the well being of an individual, the nation as well as the economic growth of Sri Lanka.

5. Preparing the people for the changes that occur in a rapidly changing world by adapting to it and controlling them; developing abilities and potentialities of people to face the complex and unexpected occasions. Sustaining the skills and attitudes based on justice, equality, mutual respect which is essential to achieve a respectable place in the international community.

   National Education Commission Report (2003).

3. Basic Competencies

The competencies promoted through the education mentioned below might help to achieve the above mentioned National Goals.

(i.) Competencies in Communication

This first set of competencies is made up of four subsets - Literacy, Numeracy, Graphics and information communication skills

- **Literacy** : Carefully listening, Speaking clearly, Reading for comprehension, writing clearly and accurately.
- **Numeracy:** Using numbers to count, calculate, code and to measure, matter, space and time.
- **Graphics:** Making sense of line and form, expressing and recording essential data, instructions and ideas with line, form, colour, two and three-dimensional configurations, graphic symbols and icons.
- **ICT Competencies:** Knowledge on computers, and the ability to use the information communication skills at learning or work as well as in the private life.

(ii.)Competencies relating to the Personality Development

- Generic skills such as creativity, divergent thinking, initiative, decision making, problem-solving, critical and analytical thinking, team work, inter-personal relationships, discovering and exploring.
- Values such as integrity, tolerance and respect for human dignity.
- Cognition.

(iii.) Competencies relating to the Environment.

This is the second set of competencies related to the Social, Biological and Physical Environments.

- **Social Environment**: Awareness, sensitivity and skills linked to being a member of society, social relationship, personal conduct, general and legal conventions, rights, responsibilities, duties and obligations.
- **Biological Environment**: Awareness, sensitivity and skills linked to the living world, man and the ecosystem, the trees, forests, seas, water, air and life - plant, animal and human life.

- **Physical Environment**: Awareness, sensitivity and skills relating to space, energy, fuels, matter, materials and their linkswith human living, food, clothing, shelter, health, comfort, respiration, sleep, relaxation, rest, wastes and excretion, media of communication and transport.
  Included here are the skills in using tools to shape and for materials for living and learning.

(iv.) Competencies relating to Preparation for the world of work

Employment-related skills to maximize their potential and to enhance their capacity to contribute to economic development; to discover their vocational interests and aptitudes; to choose a job that suits their abilities and; to engage in a rewarding and sustainable livelihood.

(v.) Competencies relating to religion and ethics

This fourth set of competencies is laden with values and attitudes. It is essential for individuals to assimilate values, so that they may function in a manner consistent with the ethical, moral and religious modes of conduct, rituals, practices in everyday living, selecting the most appropriate.

(vi.) Competencies in Play and Use of Leisure

Competencies that link up with pleasure, joy, emotions and such human motivations. These find expression in play, sports, athletics and leisure pursuit of many types. These also link up with such values as cooperation, team work, healthy competition in life and work. Here are included such activities as are involved in aesthetics, arts, drama, literature, exploratory research and other creative modes in human living.

(vii.) Competencies relating to 'Learning to learn'.

These competencies flow directly from the nature of a rapidly changing, complex and interdependent and crowded world. Whatever one learns, that learning will need updating and review. This requires that one should be aware of, sensitive and skilful in sustained attention, and be willing to persevere and attend to details that matter in a given situation.

National Education Commission Report (2003).

4. Aims of the Information and Communication Technology Curriculum

Such a surge in the growth, development and the application of Information and Communication Technology as today has never been experienced before. The importance and relevance of ICT to almost all walks of life today has made it all the more important that knowledge and expertise, both practical and theoretical, of its application, should begin at the very grass roots level of education.

Aims to be achieved by the course are as follows:

- Inculcate basic computer literacy and develop a base for further pursuit of Information Technology and Communication Technology studies.
- Develop understanding of use and resultant outcomes of use of different types of ICT applications.
- Develop concepts and principles related to ICT.
- Improve skills required for the development of ICT based solutions for real world problems.
- Provide awareness of benefits and problems of ICT use to participants

5. Syllabus of G.C.E. (O/L) ICT: Grade 10

Subject Content and Time Duration by Competencies and Competency Levels

Competency	Competency Level	Content	Learning Outcomes	Periods
1. Investigates the place of the computer in the world of information.	**1.1 Investigates the contribution of ICT towards national development.**	• Introduction to ICT • Application of ICT in the society o e-government, agriculture, education, health, industry, entertainment, defence, media, communication, social networking, environment	• Explains the nature of ICT • Elaborates on the uses of ICT in various fields of work	3
	1.2 Investigates the Computer as a system for converting data into information.	• Components of a system. • Difference between data and information. • Data as input and information as output • Processing as the method for converting data into information.	• Describes various systems in day-to-day life • Elaborates on the functions of an information system in terms of its main components.	2

Competency	Competency Level	Content	Learning Outcomes	Periods
	1.3 Explores the evolution of computers to identify their major developments.	• Computer generations • Evolution of processor technology ○ Vacuum tubes ○ Transistors ○ Integrated circuits: LSI,VLSI, Quantum Computing • Improvements in system characteristics ○ Size ○ Capacity ○ Speed ○ Accuracy ○ Efficiency	• Explains landmarks in the evolution of computers • Discusses the enhancement in system characteristics with the evolution.	2
2. Selects and uses Computer Hardware.	**2.1 Classifies computers using a variety of methods.**	• Computer classifications ○ Size: Super, Main frame, Mini, Micro ○ Technology: Digital, analog and hybrid	• Briefly explains computer classification.	1
	2.2 Explores the computer systems by function.	• Functions of a computer and its peripherals ○ Input ○ Processing ○ Output ○ Storage ○ Communication	• Identifies components of a computer system • Describes the functions of each component	3

Competency	Competency Level	Content	Learning Outcomes	Periods
	2.3 Identifies and connects basic peripherals to the computer.	• Basic computer components: keyboard, mouse, system unit, and monitor • Ports o PS/2, Serial, Parallel ports (Obsolete) o USB port o Ethernet port o Video port (VGA, HDMI, etc..) o Audio port	• Describes basic components of a computer • Describes the use of computer ports	2
	2.4 Uses the basic block diagram to demonstrate the computer system.	• CPU: Control Unit ALU and Memory Registers • Flow of signals o Control signals o Data and instruction signals	• Identifies components of the CPU in a block diagram. • Explains the flow of control, data, and instruction signals.	2

Competency	Competency Level	Content	Learning Outcomes	Periods
	2.5 Investigates the benefits and concerns of computer networks for optimal communication.	• Purpose of computer networks ○ Data communication ○ Resource sharing • Components of a network ○ Networking devices (NIC, Modem, Switch /Hub, Router, Firewall) ○ Transmission media: wired and wireless ▪ Wired: Network cables: twisted pair, coaxial, fiber optics ▪ Wireless: Radio waves, Infrared, Microwave Switch, router • Network operating system • Types of computer networks ○ LAN ○ WAN ○ PAN • Advantages and disadvantages of networks	• Explains the purpose of networking • Describes the physical layout of a network • Discusses the issues in networking. • Distinguishes between wired and wireless transmission media • Explains the role of a Network Operating System (NOS) • Classifies different types of computer networks, with real-life examples. • Discusses the advantages and disadvantages of computer networks	3

Competency	Competency Level	Content	Learning Outcomes	Periods
3. Investigates the methods used to represent data in computer systems.	**3.1. Uses the binary number system to represent data in computer systems.**	• Different methods of data representation ○ Two-state systems (Using 1 and 0) ○ Multi-Stage Systems (Quantum Computing) • Binary number system	• Explains that data can be represented using two states • Represents positive decimal integers in binary	2
	3.2. Uses different number systems and converts decimal numbers to binary, octal, and hexadecimal.	• Number systems ○ Decimal, Binary, Octal, and Hexadecimal • Need of Number Systems	• Describes number systems with 2, 8,10, and 16 as the bases • Computes position values of the numbers converted in each system. • Converts decimal numbers between given number systems	2
	3.3. Uses different number systems and performs conversions between number systems.	• Methods for conversions (positive integers only): ○ Binary to Decimal ○ Octal to Decimal ○ Hexadecimal to Decimal ○ Binary to Octal ○ Octal to Binary ○ Binary to Hexadecimal ○ Hexadecimal to Binary ○ Octal to Hexadecimal ○ Hexadecimal to Octal	• Converts binary, octal, and hexadecimal integers to decimals • Converts binary integers to octals and vice versa • Converts binary integers to hexadecimal and vice versa • Converts octal integers to hexadecimal and vice versa	4

Competency	Competency Level	Content	Learning Outcomes	Periods
	3.4. Determines the capacity of data storage	• Units of measurement: Bit, Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte • Order of capacities of different storage devices: Registers, Cache, RAM ROM,CD,DVD, Blu-Ray, Flash, Hard Disk , SSD • Cloud Storage	• Describes storage units in terms of bytes • Compares the capacity of various storage devices	2
	3.5. Explores coding systems in computers	• Different Coding Systems for Data Representation o BCD ,ASCII, Unicode	• Explains how different coding systems are used • Explains limitations of each system	1
4. Uses Boolean logic to work effectively with logic gates	**4.1 Identifies basic Logic Gates and draws truth tables to illustrate their functions**	• Logic Gates: AND, OR ,NOT • Introduction to Truth Tables (maximum of three inputs)	• Explains the action of logic gates • Draws truth tables for logic operations	2
	4.2 Applies concepts of Boolean logic to find solutions to simple day-to-day life problems.	• Universal Logic Gates – (NAND, NOR) • Design of logic for simple real-world applications. o Alarm systems o Street Light	• Draws block diagrams of systems using Boolean logic • Draws block diagrams to represent solutions to simple problems involving Boolean logic • Converts block diagrams into logic diagrams	2

Competency	Competency Level	Content	Learning Outcomes	Periods
5. Works effectively with Operating Systems	**5.1 Explores operating systems by type, functions, benefits and concerns.**	• Introduction to OS • Evolution of computer/human interface • Customization of the OS Environment • Types of operating systems: single user, multi-user, multi tasking, real-time • Functions of the OS: user interface and resource management • Execution of Application software • Benefits of the OS • Utilities of an OS: o Partitioning, o Formatting, o Defragmentation / Optimize drives	• Describes the need of an operating system • Explains functions of an operating system • Explains the utilities of an operating system	4
	5.2 Handles files and folders in Operating Systems	• Introduction to the files organization o Drives o Folders o Files and file extensions o File and folder operations	• Carries out the following operations: creation, deletion, renaming, copying • Organizes documents into folders according to to need using different user interfaces	2

Competency	Competency Level	Content	Learning Outcomes	Periods
6. Uses Wordprocessing Software to Solve Day-to-day problems	**6.1 Explores the concept and features of wordprocessing**	• Functions of word-processing software • GUI of wordprocessing software	• Discusses the facilities of Wordprocessing software • Explores the GUI of Wordprocessing software	2
	6.2 Performs basic tasks in the Wordprocessing software	• Creating new documents • Opening existing documents • Saving and closing of documents	• Creates new documents • Opens existing documents Saves and closes documents	2
	6.3 Uses different types of formatting in word processing	• Formatting text • Graphics: Insertion and formatting • Shapes: Insertion and formatting	• Applies suitable text formatting, manipulates graphics, and draws simple shapes	2
	6.4 Manipulates table feature in wordprocessing software	• Insertion of tables • Column width and height • Deletion and insertion of columns and rows • Splitting and merging of cells	• Creates tables to insert Data Format and Edit Tables	2
	6.5 Creates and prints documents	• Spelling and grammar checking • Find and replace text • Page setup: paper size, margins, orientation and selection of a printer • Print options: copies and page range.	• Checks spelling and grammar of a document • Prints documents with necessary settings	2
	6.6 Uses the Mail Merge facility	• Document types o Common document o Source data	• Creates template document • Creates data tables • Merge data with template	3

Competency	Competency Level	Content	Learning Outcomes	Periods
7. Uses a spreadsheet to solve simple statistical problems	**7.1. Explores a Spreadsheet to identify its basic features and functions.**	• Introduction of spreadsheets • GUI of the spreadsheet software	• Lists the functions of an electronic spreadsheet • Identifies features of GUI of a spreadsheet Software	1
	7.2. Moves around the worksheet to gain hands-on experience of data entry	• Worksheet, columns, rows and cells • Moving around worksheet. Data entry (label, number, formulae)	• Identifies components of a worksheet Enters and edits data	1
	7.3. Performs basic Mathematical operations	• Simple calculations using values and operators (+, -, *, /,^) • Simple calculations using cell addresses and operators (+, -, *, /, ^)	• Carries out calculations using basic operators • Uses cell references in calculations.	1
	7.4. Uses built-in functions for calculations	• SUM, AVERAGE, MAX, MIN, COUNT, COUNTA, ROUND	• Identifies functions and its parameters • Uses basic built-in functions in calculations	3
	7.5. Applies conditions and organizes data using sorting and filtering	• IF, IF-ELSE, Sorting, filtering	• Applies If and If Else formulas in spreadsheets • Sorts data in ascending and descending order • Filters data to show specific information • Combining conditions with sorting and filtering	3

Competency	Competency Level	Content	Learning Outcomes	Periods
	7.6 Formats a worksheet	• Font, Font size, Boldface, Italic. • Alignment (centre, left, right, justify) • Decimal places (increase and decrease) • Merge and Center	• Formats and aligns data in cells • Sets decimal places according to requirements	1
	7.7 Uses relative versus absolute cell references appropriately.	• Absolute and Relative cell References • RANK function	• Explains relative and absolute addressing modes • Uses both modes appropriately in calculations • Use RANK function to orderly arrange data	3
	7.8 Creates charts using a spreadsheet to explain data	• Charts: chart type, chart options • Chart types: Bar, column, line, and pie	• Identifies the relevant chart type • Present data using suitably built-in chart types	2
8. Uses Presentation software to develop electronic presentations.	**8.1 Produces effective Presentations integrating multimedia.**	• Introduction to presentation software • Changing Background, Slide layout, Slide designs. • Inserting text and multimedia	• Applies good practices in the use of presentation software • Formats slide layout Inserts text, images, movies and sounds	3

Competency	Competency Level	Content	Learning Outcomes	Periods
	8.2 Applies suitable animations to enhance the quality of presentations.	• Slide transitions • Custom animation • Slide Master • Slide show	• Applies suitable screen transitions • Applies suitable animations on screen objects. Record a slide show after formatting	3
9. Develops simple databases to elicit information.	**9.1 Explores the Concept of Database**	• Introduction to databases • Difference between a dataset and a database • Need of different types of databases –e.g., NoSQL • Advantages of databases. • Features of a well-designed database: - Minimize of redundancy, Efficiency, Accuracy, Consistency, Security, Validity, Simplicity, Integrity • Manual and electronic Databases. • Difference between Electronic and manual databases • Introduction to Relational Databases: Tables, records, fields, key fields	• Discusses nature and advantages of databases • Explains the features of relational databases	3

Competency	Competency Level	Content	Learning Outcomes	Periods
	9.2 Creates a simple A database with a single table, manually.	• Field name, unique field, data types, field size	• Identifies field names, data types, and unique fields in a database. Creates a simple data table using appropriate fields and data types.	2
	9.3 Converts a manual database into electronic media.	• Field name, unique field, data types, field size, field Properties	• Identifies the purpose of the database • Select suitable fields to create data tables	2
	9.4. Uses relational database concepts to design a simple relational database.	• Tables, fields and key fields: Primary key, foreign key, relationships	• Designs simple relational databases manually • Identifies primary and foreign keys	2
	9.5 Uses DBMS software to create relational databases	• Creation of a database • Identification of primary and foreign key • Creation of a relationship between tables	• Creates a database using DBMS software with multiple tables. • Establishes relationships using primary and foreign keys.	2
	9.6 Uses forms to view and update data	• Form design • Manipulation of properties of a form • Insertion of controls: Delete button	• Designs data input and editing forms • Inserts suitable controls to manipulate data	2

Competency	Competency Level	Content	Learning Outcomes	Periods
	9.7 Creates Queries to extract information	• Design of queries using the query tool without using SQL structure • Use of criteria • Sorting of records	• Design queries to meet given criteria Performs simple queries on the database	2
	9.8 Creates reports to present information	• Use of the report Wizard • Printing of reports	• Creates reports for given purposes • Prints reports	2
			Total(Periods)	90

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Competency: 1.0 Investigates the place of the computer in the world of information.
Competency Level: 1.1 Investigates the contribution of ICT towards national development

Time: 3 periods

Learning Outcomes:

- Explains the nature of ICT
- Elaborates the uses of ICT in various fields of work

Content:

- Introduction to ICT
- Application of ICT in society :
  e-government, Agriculture, Education, Health, Industry, Entertainment, Defence, Communication media, Information Communication, Social Media, Environment

Concepts and terms need to be highlighted:

- Introduction to ICT
- Brief introduction to fields in which ICT is used with real-world examples

Guidance for lesson plans:

Students are supposed to explore the use of ICT and participate in a classroom discussion.

Guidance for assessments and evaluations:

Preparation of a booklet on the use of ICT in the given fields.

Quality inputs:

- https://medicalcaremedia.com/the-importance-of-ict-in-healthcare-how-technologyis-saving-lives/

Competency Level: 1.2 Investigates the computer as a system for converting data into information.

Time: 2 periods

Learning Outcomes:

- Describes various systems in day-to-day life
- Elaborates on the functions of an information system in terms of its main components

Content:

- Components of a system
- Difference between data and information
- Data as input and information as output
- Processing as the method for converting data into information

Concepts and terms need to be highlighted:

- Ensure that students clearly identify data, processing and information.
- Ensure that students can distinguish between data and information, emphasizing that organized data is considered information.
- Stress that data and information are relative terms and they can be better identified with respect to a data processing system Information in a certain sense could be data with respect to some other system.

Guidance for lesson plans:

Discuss with numerous examples.

Guidance for assessments and evaluations

1. Provide facts to the students (students with items) and let them decide whether they are data or information.
2. Ask the logic behind their decisions.
3. Provide any two items out of input, process and output and let students decide the missing part with reasoning.

Quality inputs:

- https://www.geeksforgeeks.org/difference-between-information-and-data/
- https://www.techtarget.com/searchdatamanagement/definition/information

Competency level: 1.3 Explores the evolution of computers to identify its major developments.

Time: 2 periods

Learning Outcomes:

- Explains landmarks in the evolution of computers
- Discusses the enhancement in system characteristics with the evolution

Content:

- Computer generations
- Evolution of processor technology
    - Vacuum tubes
    - Transistors
    - Integrated circuits (LSIC,VLSI, Quantum Computing)
- Improvements in system characteristics
    - Size
    - Capacity
    - Speed
    - Accuracy
    - Efficiency

Concepts and terms need to be highlighted:

Emphasize the improvement in the processor capacity and efficiency with respect to the evolution.

Guidance for lesson plans:

1. Design a group activity.
2. Allow students to search for information on the evolution and create a booklet.
3. Get the students make presentations on their discoveries.

Guidance for assessments and evaluations

Set MCQs covering the main points of this evolution.

Quality inputs:

- https://history.computer.org/pubs/timeline.pdf
- https://www.slideshare.net/slideshow/evolution-of-the-computers/3775322
- https://www.ibm.com/think/topics/central-processing-unit-history

Competency : 2.0 Selects and uses computer hardware.
Competency Level: 2.1 Classifies computers using a variety of methods.
2.2 Explores computer systems by function.
Time: 4 periods

Learning Outcomes:

- Identifies different types of computers based on size.
- Classifies computers according to the technology used.
- Identifies components of a computer system
- Describes the functions of each component

Content:

- Functions of a computer and its peripherals
    - Input
    - Output
    - Processing
    - Storage
    - Communication

Concepts and terms need to be highlighted:

- Describing the peripherals with their functions (keyboard, mouse, touch screen, light pen, digitizer, microphone, webcam, OCR, MICR, OMR, monitor, printer, plotter, speaker)
- Selection of peripherals according to requirements
- Compare and contrast storage devices in terms of capacity, speed and relative position in the storage hierarchy

Guidance for lesson plans:

1. Physical display of peripherals
2. Use movies to describe functionality of uncommon peripherals
3. Analyse computer system advertisements with specifications from newspapers or online sources to understand component specifications and purposes.

Guidance for assessments and evaluations:

Provide students with requirement specifications to purchase a computer and ask them to specify hardware for that purchase.

Quality inputs:

- https://www.geeksforgeeks.org/input-and-output-devices/
- https://cstutorialpoint.com/computer-fundamentals-tutorial/
- https://www.chtips.com/computer-fundamentals/types-of-scanners/

Competency level: 2.3 Identifies and connects basic peripherals to the computer.

Time: 2 periods

Learning Outcomes:

- Identifies basic computer components and types of ports
- Connects peripherals to suitable ports on a computer

Content:

- Basic computer components: keyboard, mouse, system unit and monitor
- Ports
    - PS/2, Serial, Parallel ports (Obsolete)
    - USB port
    - Ethernet port
    - Video port (VGA, HDMI, etc.)
    - Audio port

Concepts and terms need to be highlighted:

- Physical identification of the ports
- Type of peripherals to be attached to each ports

Guidance for lesson plans:

1. Design group activities.
2. Let each group find out various peripherals and the relevant ports of them.

Guidance for assessments and evaluations

Provide exercises to map between peripheral devices and ports.

Quality inputs:

- https://www.tutorialspoint.com/computer_fundamentals/computer_ports.htm
- https://www.geeksforgeeks.org/universal-serial-bus-usb/
- https://www.etechnophiles.com/different-types-of-computer-ports/

Competency level: 2.4 Uses the basic block diagram to demonstrate the computer system

Time: 2 periods

Learning Outcomes:

- Identifies components of the CPU in a block diagram.
- Explains the flow of control, data, and instruction signals

Content:

- Central Processing Unit
  - Control Unit, ALU and Memory Registers
- Flow of signals
  - Data and instruction signals
  - Control signals

Concepts and terms need to be highlighted:

Brief explanation on data, instructions and controls

Guidance for lesson plans:

1. Discussion.
2. Use a motherboard to show the paths.
3. Use animations to explain the signal flow.

Guidance for assessments and evaluations

Provide MCQ test based on the demonstration of the flow of signals.

Quality inputs:

- https://www.geeksforgeeks.org/central-processing-unit-cpu/
- https://www.tutorialspoint.com/computer_fundamentals/computer_motherboard.htm

Competency level: 2.5 Investigates benefits and concerns of computer networks for optimal communication.

Time: 3 periods

Learning Outcomes:

- Explains the purpose of networking
- Describes physical layout of a network
- Discusses the issues in networking
- Distinguishes between wired and wireless transmission media
- Explains the role of a Network Operating System (NOS)
- Classifies different types of computer networks, with real-life examples
- Discusses the advantages and challenges of computer networks

Content:

- Purpose of computer networks
    - Data communication
    - Resource sharing
- Components of a network
    - Networking devices (NIC, Modem, Switch /Hub, Router, Firewall)
    - Transmission media: wired and wireless
        - Wired: Network cables: twisted pair, coaxial, fiber optics
        - Wireless: Radio waves, Infrared, Microwave
- Network operating system
- Types of computer networks
    - Local Area Network - LAN
    - Wide Area Network - WAN
    - Personal Area Network - PAN
- Advantages and challenges of networks

Concepts and terms need to be highlighted:

Enough real-time examples of computer networks in various organizations: banks, supermarkets, office.

Guidance for lesson plans:

1. Use network available in the computer lab to explain network characteristics of a Local Area Network.
2. Physical observation of a network of an institution.
3. Demonstrate with an animation.

Guidance for assessments and evaluations:

Give students simple LAN specifications and ask them to create a rough diagram of the network, describing the components used.

Quality inputs:

- https://www.geeksforgeeks.org/basics-computer-networking/
- https://www.cloudflare.com/learning/network-layer/what-is-a-personal-area-network/

Competency: 3.0 Investigates the methods used to represent data in computer systems.
Competency level: 3.1 Uses the binary number system to represent data in computer systems.
Time: 2 periods

Learning Outcomes:

- Explains that data can be represented using two states
- Represents positive decimal integers in binary

Content:

- Different methods of data representation
    - Two-state systems (Using 1 and 0)
    - Multi-state systems (Quantum computing)

- Binary number system

Concepts and terms need to be highlighted:

- Concept of two states.
- Various types of data: Text, numbers, images, audio, video.
- Representation of data in relevant media

Guidance for lesson plans:

1. Explain the concept of two states with respect to digital data.
2. Discuss various types of data: Text, numbers, Images, Audio, video
3. Discuss how they are represented in relevant media.
4. Demonstrate how data is represented in various media using visual aids.
5. Use base 10 and base 2 to represent positive integers.
6. Guide students to explore the use of different number systems in day-to-day life.

Guidance for assessments and evaluations:

1. Give exercises to the students on conversion of decimal positive integers to binary and vice versa.
2. Instruct students to prepare a presentation including the use of different number systems in day-to-day life.

Quality inputs:

- https://www.cs.princeton.edu/courses/archive/spr19/cos217/lectures/05_NumberSystems.pdf
- https://www.geeksforgeeks.org/introduction-quantum-computing/

Competency Level: 3.2 Uses different number systems and converts decimal numbers to binary, octal, and hexadecimal.

Time: 2 periods

Learning Outcomes:

- Describes number systems with 2, 8 ,10 and 16 as the bases
- Computes position values of the numbers converted in each system
- Converts decimal numbers between given number systems

Content:

- Number systems
    - Decimal, Binary, Octal, and Hexadecimal
- Need of Number Systems

Concepts and terms need to be highlighted:

- Focus only on positive integers.
- Explaining with examples where these number systems are used. (RGB: Hexadecimal)

Guidance for lesson plans:

Demonstrate number conversions with examples.

Guidance for assessments and evaluations:

Guide students through hands-on activities for conversions.

Quality inputs:

- https://www.geeksforgeeks.org/how-many-types-of-number-systems-are-there/
- https://www.tutorialspoint.com/computer_fundamentals/computer_number_system.htm

Competency Level: 3.3 Conversion between different number systems.

Time: 4 periods

Learning Outcomes:

- Converts Binary , Octal and hexadecimal integers to decimals
- Converts binary integers to octals and vice versa
- Converts binary integers to hexadecimal and vice versa
- Converts octal integers to hexadecimal and vice versa

Content:

- Methods for conversions (positive integers only):
  - Binary to Decimal
  - Octal to Decimal
  - Hexadecimal to Decimal
  - Binary to Octal
  - Octal to Binary
  - Binary to Hexadecimal
  - Hexadecimal to Binary
  - Octal to Hexadecimal
  - Hexadecimal to Octal

Concepts and terms need to be highlighted:

Focus only on positive integers.

Guidance for lesson plans:

1. Demonstration of conversions.
2. Get students to understand the rationale behind conversions between octal and binary and hexadecimal and binary.

Guidance for assessments and evaluation:

Guide students through hands-on activities for conversions.

Quality inputs:

- https://www.geeksforgeeks.org/number-system-and-base-conversions/
- https://www.tutorialspoint.com/octal-number-system
- https://teachcomputerscience.com/uses-of-hexadecimal/

Competency Level: 3.4 Determines the capacity of data storage.

Time: 2 periods

Learning Outcomes:

- Describes storage units in terms of bytes
- Compares the capacity of various storage devices

Content:

- Units of measurement
  Bit, Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte
- Order of capacities of different storage devices:
  Registers, Cache, RAM, ROM, CD, DVD, Blu-ray, Flash drive, Hard disk, SSD
- Cloud Storage

Concepts and terms need to be highlighted:

- Logic behind the existence of many units
- Order of capacities of common storage devices

Guidance for lesson plans:

1. Display a variety of storage devices.
2. Ask to find out capacities.
3. Exercises on conversion between units.

Guidance for assessments and evaluations:

Guide students to find the capacities of various storage devices of a given computer.

Quality inputs:

- https://www.geeksforgeeks.org/storage-systems-in-cloud/
- http://www.tutorialspoint.com/computer_fundamentals/computer_memory_units.htm

Competency Level: **3.5 Explores coding systems in computes.**

Time: 1 period

Learning Outcomes:

- Explains how different coding systems are used
- Explains limitations of each system

Content:

- Different Coding systems for Data Representation
    - BCD
    - ASCII
    - Unicode

Concepts and terms need to be highlighted:

Use of Unicode

Guidance for lesson plans:

1. Explain ASCII code with word processing (eg. type ALT + 65 from number key pad for letter 'A')
2. Explain the need of Unicode.
3. Explain why some coding systems are used less frequently at present.

Guidance for assessments and evaluations:

1. Provide exercises on the conversion of decimal numbers to BCD.
2. Provide exercises on the conversion of ASCII with fewer than 3 characters.

Quality inputs:

- https://www.electronics-tutorials.ws/binary/binary-coded-decimal.html
- https://www.geeksforgeeks.org/what-is-ascii-a-complete-guide-to-generating-ascii-code/
- https://www.unicode.org/standard/WhatIsUnicode.html

Competency: 4.0 Uses Boolean logic to work effectively with logic gates.

Competency Level: **4.1 Identifies basic logic operators and draws truth tables to illustrate their functions.**

Time: 2 periods

Learning Outcomes:

- Explains the action of logic gates
- Draws truth tables for logic operations

Content:

- Logic Gates: AND, OR, NOT
- Introduction to Truth Tables (maximum of three inputs)

Concepts and terms need to be highlighted:

Brief discussion of the use of gates in computer hardware

Guidance for lesson plans:

1. Explanation of the concept of gates using day-to-day examples.
2. Simulation of gates using electrical circuits.
3. Demonstrate the output of a combinational circuit using a software.(MMLogic)

Guidance for assessments and evaluations:

1. Guide students with hands-on activities to construct truth tables for given circuits.
2. Encourage designing a circuit using a two-input truth table.

Quality inputs:

- https://www.geeksforgeeks.org/logic-gates/
- https://computer.howstuffworks.com/boolean.htm
  https://multimedia-logic.software.informer.com/1.6/

Competency Level: 4.2 Applies concepts of Boolean logic to find solutions to simple day-to- day life problems.

Time: 2 periods

Learning Outcomes:

- Draws block diagrams of systems using Boolean logic
- Draws block diagrams to represent solutions to simple problems involving Boolean logic
- Converts block diagrams into logic diagrams

Content:

- Universal Logic Gates – (NAND, NOR)
- Design of logic for simple real-world applications
    - Alarm systems
    - Street Light

Concepts and terms need to be highlighted:

Discussion of real life problems where Boolean logic is applied.

Guidance for lesson plans:

Group activity:

1. Use day-to-day life problems relating to warning systems or control systems.
2. Guide students to design the block diagram that shows the Boolean logic behind the solution.
3. Let each group present its solution to the class.

Guidance for assessments and evaluations:

Provide exercises to present solutions to problems that should be addressed using Boolean logic through block diagrams.

Quality inputs:

- https://www.tutorialspoint.com/electronic_circuits/index.htm

Competency: 5. 0 Works effectively with Operating Systems
Competency Level: 5.1 Explores operating systems by type, functions, benefits and concerns.
Time: 4 periods

Learning Outcomes:

- Describes the need of an operating system
- Explains functions of an operating system
- Explains the utilities of an operating system

Content:

- Introduction to OS
- Evolution of computer/human interface
- Customization of the OS Environment
- Types of operating systems: single-user, multi-user, multi-tasking, real-time
- Functions of the OS: user interface and resource management,
- Execution of Application software
- Benefits of the OS
- Utilities of an OS: Partitioning, Formatting, Defragmentation/Optimizing drives

Concepts and terms need to be highlighted:

- Evolution of Human Computer Interface: Command prompt, GUI, Touch screen, Voice commands
- Examples of types of operating systems
- Basic operations such as control of hardware, storage management, process management, error handling, user interface
- Customization of the operating system using Display settings, Personalization

Guidance for lesson plans:

Group Activity:

1. Divide the classroom into groups.
2. Assign one task to each group: Types of operating systems, functions, benefits and utilities.
3. Encourage one member of each group to make a presentation of the finding of the group.
4. Teacher is supposed to elaborate on their findings and complete the lesson.
5. Demonstrate partitioning, formatting and defragmentation of a hard disk.

Guidance for assessments and evaluations:

1. Give several cases and ask students to suggest a suitable operating system to be used.
2. Guide students to list out the functions of a network operating system.
3. provide hands on activities to illustrate the functional examples of the operating system within these structures.

Quality inputs:

- https://www.tutorialspoint.com/operating_system/os_overview.htm
- https://www.techtarget.com/searchnetworking/definition/network-operating-system

Competency Level: 5.2 Handles files and folders in Operating Systems.

Time: 2 periods

Learning Outcomes:

- Carries out the following operations: creation, deletion, renaming, copying
- Organizes documents into folders according to needs using different user interfaces

Content:

- Introduction to the files organization
    - Drives
    - Folders
    - Files and file extensions
    - File and folder operations

Concepts and terms need to be highlighted:

- User interface of the Operating system being used
- Use of operating system facilities available
- Customization of user interface: desktop, shortcuts, screen saver, power saving
- Managing folders and files according needs of work organization

Guidance for lesson plans:

1. Hands-on use of the operating system in the computer lab.
2. Provide worksheets to try out operating system operations.

Guidance for assessments and evaluations:

1. Conduct practical assessments to check operating system functions.
2. Provide a list of user requirements for a document management activity taking place in an office and guide students to create suitable folder structure for it.

Quality inputs:

- https://www.tutorialspoint.com/what-are-the-functions-of-the-operating-system
- https://www.geeksforgeeks.org/structures-of-directory-in-operating-system/

Competency: 6.0 Uses Word-processing Software to Solve Day-to-day problems
Competency Level: 6.1 Explores the concept and features of word processing
6.2 Performs basic tasks in word processing software

Time: 4 periods

Learning Outcomes:

- Discusses the facilities of wordprocessing software
- Explores GUI of word-processing software
- Creates new documents
- Opens existing documents
- Saves and closes documents

Content:

- Functions of wordprocessing software
- GUI of word-processing software
- Creating new documents
- Opening existing documents
- Saving and closing of documents

Concepts and terms need to be highlighted:

1. Functionalities available: Creation, Editing, Storing, use of graphics, Copying, Cut, Paste
2. Familiarization of the user interface.
3. Short-cuts for menu sequences: to open, close, save, copy, cut, paste, undo, redo

Guidance for lesson plans:

1. Demonstrate the basic operations of the software.
3. Different ways of opening a wordprocessing document: through the software, by double-clicking the document, through right-clicking.
4. Practical demonstration of keeping a document prepared with password protection.

Guidance for assessments and evaluations:
Guide students to,

1. Opening of a saved document on the hard disk
2. Editing the document
3. Saving in given formats

Quality inputs:

- https://www.guru99.com/best-free-word-processor.html

Competency level 6.3: **Uses different types of formatting in wordprocessing.**

Time: 2 periods

Learning Outcomes:

- Applies suitable text formatting
- Manipulates graphics and draws simple shapes

Content:

- Formatting text
- Graphics: Insertion and formatting
  Shapes: Insertion and formatting

Concepts and terms need to be highlighted:

- Various types of formatting: Text (fonts, boldface, italics..), paragraph (center, right and left alignment, justification, bullets, and numbers), page (margins, orientation)

Guidance for lesson plans:

1. Teacher demonstration of the tools of formatting.
2. Provide worksheets created to cover formatting techniques.
3. Provide meaningful exercises to enable them to learn in context.
4. Use of Unicode for Sinhala/Tamil fonts.

Guidance for assessments and evaluations:

1. Provide with unformatted softcopies to the students to practice the required formatting.
2. Encourage students to do exercises for designing the completed documents using software.

Quality inputs:

- http://www.guru99.com/best-free-word-processor.html
- https://officemastery.com/text-formatting-tools-in-word/

Competency level: 6.4 Manipulates table features in wordprocessing software.

Time: 2 periods

Learning Outcomes:

- Creates tables to insert data
- Formats and Edits tables

Content:

- Insertion of tables
- Column width and height
- Deletion and insertion of columns and rows
- Splitting and merging of cells

Concepts and terms need to be highlighted:

- Adjustment of column width: exact fit
- Merging of cells as the final editing

Guidance for lesson plans:

1. Demonstrate the use of tools of the tables.
2. Provide worksheets to exercise the usage of tables.
3. Provide meaningful exercises to enable them to learn in context.

Guidance for assessments and evaluations:

1. Provide printed copies of worksheets including exercises with merged cells.
2. Guide students to use word processing software to create those tables.

Quality inputs:

- https://www.tutorialspoint.com/word/word_create_table.htm
- https://www.geeksforgeeks.org/ms-word-tutorial/

Competency level 6.5: Creates and prints documents.

Time: 2 periods.

Learning Outcomes:

- Checks spelling and grammar of a document
- Prints documents with necessary settings

Content:

- Spelling and grammar checking
- Find and replace of text
- Page setup: paper size, margins, orientation and selection of printers
- Print options: copies and page range

Concepts and terms need to be highlighted:

- Selection of the right printer when several printer software is installed
- Limitations in built-in spellchecker and grammar checker
- Effective use of the replace text facility: type an abbreviation and replace with the longer text

Guidance for lesson plans:

1. Provide documents to type with spelling mistakes.
2. Apply spell-checker to correct mistakes.
3. Use printer to take a hard-copy meeting given specifications.

Guidance for assessments and evaluations:

1. Provide documents with spelling mistakes and make them correct using spelling and grammar
2. Provide a hardcopy and let the students to identify repeating words and apply find and replace facility for fast typing.

Quality inputs:

- https://www.tutorialspoint.com/word/word_printing_documents.htm

Competency Level: 6.6 Uses the Mail Merge facility.

Time: 3 periods

Learning Outcomes:

- Creates template document
- Creates data tables
- Merges data with template

Content:

- Document types
    - Common document
    - Source data

Concepts and terms need to be highlighted:

- Usefulness of mail merge
- Familiarization with the mail merge
- Selection of a suitable type of letters

Guidance for lesson plans:

1. Provide the data table and master letter as hardcopies.
2. Ask students to create the master letter.
3. Ask students to carry out the mail merge.
4. Print the letters to a file.
5. Provide students with a letter suitable for mail merge.
6. Guide them to design the data table and master document.

Guidance for assessments and evaluations:

1. Provide data and master document in hardcopy.
2. Students are required to carry out mail merge and save in a file.

Quality inputs:

- https://www.geeksforgeeks.org/how-to-use-mail-merge-in-ms-word/
- https://www.wikihow.com/Mail-Merge-in-Microsoft-Word

Competency: 7.0 Uses a spreadsheet to solve simple statistical problems.

Competency Levels:

7.1 Explores a Spreadsheet to identify its basic features and functions.

7.2 Moves around the worksheet to gain hands-on experience of data entry

Time: 2 periods

Learning Outcomes:

- Lists functions of an electronic spreadsheet
- Identifies features of GUI of spreadsheet software
- Identifies components of a worksheet
- Enters and edits data

Content:

- Introduction of spreadsheets
- GUI of spreadsheet software
- Worksheet, columns, rows and cells
- Moving around worksheet.
- Data entry (label, number, formulae)

Concepts and terms need to be highlighted:

- The concept of spreadsheets
- Advantages of electronic spreadsheets over manual spreadsheets
- Insertion, deletion and renaming of worksheets.

Guidance for lesson plans:

1. Ask students to provide examples of tabulated data.
2. Use tabulated data for the concept of spreadsheets.
3. Demonstrate the basic documents and their features using a completed spreadsheet.
4. Provide worksheet assessments for the students to get familiar with the GUI.
5. Provide data to input into the electronic worksheet.

Guidance for assessments and evaluations:
Guide students to create worksheets with software with suitable using collection of suitable data.

Quality inputs:

- https://www.semrush.com/blog/spreadsheet-software/?msockid=0c5e87ea076868bd388c92fc06326941
- https://www.solveyourtech.com/how-to-create-a-spreadsheet-in-excel-a-step-by-step-guide/
- http://analysistabs.com/excel/introduction-to-excel-interface-and-objects/

Competency Level: 7.3 Performs basic mathematical operations.

Time: 1 period

Learning Outcomes:

- Carries out calculations using basic operators
- Uses cell references in calculations

Content:

- Simple calculations using values and operators (+, -, *, /,^)
- Simple calculations using cell addresses and operators (+, -, *, /,^)

Concepts and terms need to be highlighted:

- Use of brackets in mathematical calculations
- Importance of using cell addresses in calculations: ease of editing
- Automatic recalculation
- Copying of expressions for the same calculations

Guidance for lesson plans:
Laboratory sessions:

1. Demonstrate how to do calculations
2. Provide exercises on calculations to be done on the software

Guidance for assessments and evaluations:

1. Provide mathematical expressions to be evaluated using spreadsheet software.
2. Provide a table of data to carry out same computations.
3. Check whether students use the software properly to avoid repeating the writing of the same mathematical expressions.

Quality inputs:

- https://support.microsoft.com/en-us/office/overview-of-formulas-in-excel -ecfdc708-9162-49e8-b993-c311f47ca173

Competency Level: 7.4 Uses Inbuilt functions for calculations.

Time: 3 periods

Learning Outcomes:

- Identifies functions and its parameters
- Uses basic inbuilt functions in calculations

Content:

SUM, AVERAGE, MAX, MIN, COUNT, COUNTA, ROUND

Concepts and terms need to be highlighted:

- Difference between inbuilt functions and user-defined functions.
- Use of cell references in using the functions.
- Insertion of columns and rows in the already existing worksheet.

Guidance for lesson plans:

Laboratory sessions:

1. Demonstrate the use of functions.
2. Provide worksheets to create an electronic version of it using software.

Guidance for assessments and evaluations:

1. Provide a data set to perform specified calculations using software.
2. Exercises on manual spreadsheet calculations.

Quality inputs:

- https://ccm.net/apps-sites/software/9803-introduction-to-spreadsheet- formulas-what-is-it-and-how-to-use/#google_vignette
- https://excelx.com/formulas-functions/

Competency Level: 7.5 Applies conditions and organizes data using sorting and filtering.

Time: 3 periods

Learning Outcomes:

- Applies IF and nested IF (IF-ELSE) formulas in spreadsheets to perform logical operations
- Sorts spreadsheet data in ascending and descending order
- Uses filters to display specific information in a spreadsheet
- Applies conditional filters to display data based on given criteria

Content:

IF, IF-ELSE, Sorting, filtering

Concepts and terms need to be highlighted:

- Logical operations and conditions (greater than, less than, equal to, etc.)
- Syntax of the IF function and nested IF statements
- Difference between sorting and filtering
- Conditional filtering based on multiple criteria
- Use of cell references in logical functions
- Enhancing the readability and usability of spreadsheets through formatting and data tools

Guidance for lesson plans:

Laboratory sessions:

1. Explanation and practical demonstration of IF and nested IF functions using real-life examples.
2. Show how to sort and filter data using spreadsheet tools.
3. Guide students through hands-on activities using sample datasets.
4. Encourage peer discussion and exploration through group task

Guidance for assessments and evaluations:

1. Provide a spreadsheet task that requires applying IF, sorting, and filtering.
2. Assess the accuracy of logical functions and effectiveness of data organization.
3. Include short written questions on function syntax and conditional logic.
4. Use a rubric to evaluate practical skills, clarity, and correct application.

Quality inputs:

- https://www.wikihow.com/Use-If-Else-in-Excel
- https://www.geeksforgeeks.org/how-to-sort-data-in-excel/

Competency Level: 7.6 Formats a worksheet.

Time: 1 Period

Learning Outcomes:

- Formats and aligns data in cells
- Sets decimal places according to requirements

Content:

- Font, Font size, Boldface, Italic
- Alignment (centre, left, right, justify)
- Decimal places (increase and decrease)
- Merge and Center

Concepts and terms need to be highlighted:

Use formatting according to the requirements.

Guidance for lesson plans:

Laboratory sessions:

1. Discuss the similarity of the IDE with word-processing software.
2. Provide hands-on activities for formatting in appropriate contexts.
3. Provide worksheets to enter data and apply formatting.

Guidance for assessments and evaluations:

1. Provide an unformatted data table.
2. Ask students to create a worksheet according to the given specifications.

Quality inputs:

- https://www.techrepublic.com/article/10-excel-tips-for-creating-spreadsheets/
- https://www.exceldemy.com/learn-excel/format-cells/

Competency Level: 7.7 Uses relative versus absolute cell references appropriately.

Time: 3 periods

Learning Outcomes:

- Explains relative and absolute addressing modes
- Uses both modes appropriately in calculations
- Uses the RANK function to arrange data in order.

Content:

- Absolute and Relative cell references
- RANK function

Concepts and terms need to be highlighted:

- The difference between relative and absolute cell references
- Behaviour of cell references in copying of formulae
- Usefulness of absolute cell references

Guidance for lesson plans:

1. Guide students to understand practically by illustrating the application of relative and absolute addresses through examples.
2. Provide guidance to the students to identify the benefits of using relative cell addresses through a group activity.
3. Ask students to find out the behavior of relative and absolute cell addresses.
4. Provide sufficient exercises of the use of both relative and absolute cell addresses.

Guidance for assessments and evaluations:

1. Problems involving both modes of addressing to be solved manually.
2. Implementation of such solutions using software.

Quality inputs:

- https://www.lifewire.com/cell-references-relative-absolute-and-mixed-3123401
- https://www.geeksforgeeks.org/relative-and-absolute-cell-references-in-ms-excel/
- https://web.pdx.edu/~stipakb/CellRefs.htm

Competency Level: 7.8 Creates charts using Spreadsheets to explain data.

Time: 2 periods

Learning Outcomes:

- Identifies the relevant chart type
- Presents data using suitable built-in chart types

Content:

- Charts: chart type, chart options
- Chart types: Bar, column, line and pie

Concepts and terms need to be highlighted:

- Selection of a suitable chart type
- Selection of non-adjacent columns
- Relocation of a chart to a new sheet
- Labeling of the chart

Guidance for lesson plans:

1. Provide a structure for the students to design a chart.
2. Provide worksheets with specifications for charts.

Guidance for assessments and evaluations:

Laboratory sessions:

Design charts for the given data sets.

Quality inputs:

- https://support.google.com/docs/answer/63824?hl=en&co=GENIE.Platform%3DDesktop
- https://www.wikihow.com/Create-a-Graph-in-Excel
- https://www.geeksforgeeks.org/types-of-charts-in-excel/

Competency 8.0 :Uses Presentation software to develop electronic presentations.
Competency Level: 8.1 Produces effective presentations integrating multimedia
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Time: 3 periods

Learning Outcomes:

- Applies good practices in the use of presentation software
- Formats slide layouts
- Inserts text, images, movies and sounds

Content:

- Introduction to presentation software
- Changing background, slide layout, slide designs
- Inserting text and multimedia

Concepts and terms need to be highlighted:

- Significance of designing the presentation on paper before use of software
- Meaningful use of multimedia

Guidance for lesson plans:

1. Provide a sample presentation and let students to draw the slide design on a paper.
2. Let students select a theme of their own and design the presentation on a paper.
3. Guide students use presentation software to convert the designs into presentations.
4. Organise a pair work to design presentations independently. Then ask them to exchange designs. Then one member has to develop the other one's design into the final product.

Guidance for assessments and evaluations

1. Provide students with a design.
2. Encourage them to create the presentation as per the design.

Quality inputs:

- https://www.storydoc.com/blog/how-to-make-a-multimedia-presentation

Competency Level: 8.2 Applies suitable animations to enhance the quality of presentations.

Time: 3 periods

Learning Outcomes:

- Applies suitable screen transitions
- Applies suitable animations on screen objects
- Records a slide show after formatting

Content:

- Slide transitions
- Custom animations
- Slide Master
- Slide show

Concepts and terms need to be highlighted:

1. Meaningful use of slide transitions.
2. Meaningful use of object animations.
3. Use of custom animations for simple simulations.

Guidance for lesson plans:

1. Provide design specifications for a presentation involving animations.
2. Guide students to convert specifications into presentations.
3. Encourage students to develop presentations with animations.

Guidance for assessments and evaluations:

Guide students to create a presentation using multimedia and animations, which has educational value.

Quality inputs:

- http://24slides.com/presentbetter/slide-transitions-animations
- https://slidemodel.com/how-to-work-with-slide-master-in-powerpoint/

Competency: **9.0 Develops simple databases to elicit information.**
Competency Level: 9.1 Develops simple databases to elicit information.

Time: 3 periods

Learning Outcomes:

- Discusses the nature and advantages of databases
- Explains the features of relational databases

Content:

- Introduction to databases
- Difference between a dataset and a database
- Need of different types of databases
- Advantages of databases.
- Features of a well-designed database:
  Minimize of redundancy, Efficiency, Accuracy, Consistency, Security, Validity, Simplicity, integrity
- Difference between Electronic and manual databases
- Introduction to Relational Databases: Tables, records, fields, key fields

Concepts and terms need to be highlighted:

- Identification of databases with reference to a theme
- Explain with examples features a database should have: Absence of redundancy, Efficiency, Accuracy, Consistency, Security, Validity, Simplicity.
- Explain why at least two tables are required to have the above qualities
- Provide examples to explain (Biodata and marks)
- Do not refer to any electronic media

Guidance for lesson plans:

1. Lead a discussion to identify places where data are stored.
2. Discuss how the data are organized. (eg. Student registration details and attendance register)
3. Provide examples to demonstrate the absence of database qualities could affect negatively.

Guidance for assessments and evaluations:

1. Give a case of a database with entities so that students can easily identify tables, fields and key fields.
2. Guide students to identify instances of related tables.
3. Direct students to find out how to establish characteristics in a database.

Quality inputs:

- https://www.geeksforgeeks.org/what-is-database/
- https://www.tutorialspoint.com/ms_access/ms_access_create_database.htm

Competency Level: 9.2 Creates a simple database with a single table, manually.

Time: 2 periods

Learning outcomes:

- Identifies the purpose of the database
- Selects suitable fields to create data tables

Content:

Field name, unique field, data types, field size

Concepts and terms need to be highlighted:

- Identification of the database
- Identification of the tables, fields, type of the fields, key fields
- Identification of the relationship
- Graphical representation of the database with tables
- Not to include calculated data like age

Guidance for lesson plans:

1. Give a case where databases are involved.
2. Guide to identify one table, its fields and types.
3. Give partially worked examples to complete relational databases.
4. Discuss examples on how to break a database into tables.
5. Start off with a single table where everything is included.
6. Let students find fields and field types.
7. Let students find out key fields.

Guidance for assessments and evaluations:

1. Give a case involving several databases.
2. Ask to identify one of them.
3. Facilitate the design a table with fields, field types and sizes.

Quality inputs:

- https://www.geeksforgeeks.org/what-are-the-different-kinds-of-tables/
- https://www.tutorialspoint.com/sql/sql-data-types.htm
- https://www.databasestar.com/database-keys/

Competency Level: 9.3 Converts a manual database into electronic media.

Time: 2 periods

Learning outcomes:

- Identifies the purpose of the database
- Selects suitable fields to create data tables

Content:

Field name, unique field, data types, field size, field Properties

Concepts and terms need to be highlighted:

- Relevant features of the IDE of the database software
- Provide names for the fields and select suitable field types and size

Guidance for lesson plans:

1. Use manual databases created in 9.2 competency level
2. Create the tables.
3. Insert data.

Guidance for assessments and evaluations:

1. Provide a paragraph describing a database.
2. Give instructions to the students to sketch the database on paper.
3. Convert it into computer database using software.

Quality inputs:

- https://blog.devart.com/top-sql-database-ide.html
- https://www.tutorialspoint.com/ms_access/ms_access_create_tables.htm
- https://www.tutorialspoint.com/ms_access/ms_access_adding_data.htm

Competency Level: 9.4 Design a simple relational database.

Time: 2 periods

Learning outcomes:

- Designs simple relational databases manually
- Identifies primary and foreign keys

Content:

Tables, fields and key fields: Primary key, foreign key, relationships

Concepts and terms need to be highlighted:

- Definitions of primary and foreign keys
- Advantages of such a connection
- Examples for situations where related entities are clearly visible
- Breaking of one table into at least two to establish relationships

Guidance for lesson plans:

1. Encourage group discussion.
2. Each group is given a case to design at least two related tables.
3. Identification of primary and foreign key fields.

Guidance for assessments and evaluations:

1. Provide a case involving a relational database.
2. Ask students to create relevant tables.
3. Ask them to identify primary and foreign keys.

Quality inputs:

- https://www.geeksforgeeks.org/difference-between-primary-key-and-foreign-key/
- https://www.tutorialsteacher.com/sqlserver/tables-relations
- https://blog.devart.com/top-sql-database-ide.html

Competency Level: 9.5 Uses DBMS software to create relational databases.

Time: 2 periods

Learning outcomes:

- Creates a database using DBMS software with multiple tables
- Establishes relationships using primary and foreign keys

Content:

- Creation of a database
- Identification of primary and foreign key
- Creation of a relationship between tables

Concepts and terms need to be highlighted:

- IDE of the software
- Use of meaningful cases to create databases
- primary key with no duplicates
- Creation of the relationship with referential integrity
- Use of test data
- Deletion of data from tables to test referential integrity
- Even though its enough to limit activities to two tables, based on students' performance, it can be allowed to have more relations.

Guidance for lesson plans:

1. Use the databases created in 9.4
2. Group work is to be arranged in the computer lab.

Guidance for assessments and evaluations:

1. Use the created databases to evaluate performance.
2. Look for meaningful fields with correct data types and sizes.
3. Check primary and foreign keys.

Quality inputs:

- https://www.geeksforgeeks.org/relational-model-in-dbms/
- https://database.guide/what-is-referential-integrity/

Competency Level: 9.6 Uses forms to view and update data.

Time: **2 periods**

Learning outcomes:

- Designs data input and editing forms
- Inserts suitable controls to manipulate data

Content:

- Form design
- Manipulation of properties of a form
- Insertion of controls(INSERT, DELETE)

Concepts and terms need to be highlighted:

- Design of a layout of a form before the use of software
- Selection of suitable fonts
- Navigation with provided controls
- Editing and appending

Guidance for lesson plans:

1. Use databases created in 9.4.
2. Arrange group work
3. Entry of meaningful data.
4. Provide an exercise to practice the insert button and delete button.

Guidance for assessments and evaluations:

Provide a predesigned database to create data entry forms.

Quality inputs:

- https://www.scribd.com/document/271632967/MS-Access-2010-Tutorial-pdf
- https://www.tutorialspoint.com/ms_access/ms_access_create_form.htm
- https://www.tutorialspoint.com/ms_access/ms_access_adding_data.htm

Competency Level: 9.7 Creates Queries to extract information.

Time: 2 periods.

Learning outcomes:

- Design queries to meet given criteria
- Performs simple queries on the database

Content:

- Design of queries using the query wizard without using SQL structure.
- Use of criteria.
- Sorting of records.

Concepts and terms need to be highlighted:

- Single table with criteria
- Two related tables with criteria
- Variety of data to enable a variety of queries

Guidance for lesson plans:

1. Group work.
2. Provide a worksheet with suggested details to be retrieved using queries.

Guidance for assessments and evaluations:

1. Provide a single table with redundant data.
2. Ask to create the database with two related (one-to-many) tables.
3. Ask about various ways of information retrieval to be implemented using queries.

Quality inputs:

- https://www.tutorialspoint.com/ms_access/ms_access_query_data.htm

Competency Level: 9.8 Creates reports to present information.

Time: 2 periods

Learning outcomes

- Creates reports for given purposes
- Prints reports

Content:

- Use of report Wizard
- Printing of reports

Concepts and terms need to be highlighted:

- Reports based on queries
- Grouping of information using the primary key
- Taking the summation of the numerical field

Guidance for lesson plans:

1. Pair-work at a computer.
2. Provide a worksheet describing a database with data.
3. Ask to create queries for the given criteria.
4. Ask to create reports based on the queries.

Guidance for assessments and evaluations:

1. Provide a database with two related tables (one-to-many)
2. Give a set of report formats to create reports

Quality inputs:

- http://www.tutorialspoint.com/ms_access/ms_access_reports_basics.htm
- http://books.libreoffice.org/en/BG73/BG7301-IntroductionToBase.html